

P.5: Implementation Costs, Administrative Law, and Audit Requirements for Algorithmic Redistricting

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Abstract

A complete **bisect** redistricting run—census data fetch, graph construction, recursive bisection, analysis, and report generation—costs under \$50,000 total for a typical state, including staff time, compute, and independent audit. Census tract boundary data (40GB+ download) is available free from the Census Bureau. A 50-state congressional redistricting run completes in approximately 18 minutes of wall-clock compute time on a modern 8-core machine, at a cloud cost of under \$200. The dominant cost is staff time for quality assurance, public engagement, and legal review—not the algorithm itself. We compare these costs against two benchmarks: Iowa’s Legislative Services Agency model (\$3 million per cycle), which is the current gold standard for low-cost non-partisan redistricting, and California’s Citizens Redistricting Commission (\$35 million+ per cycle), which is the highest-cost commission model. Algorithmic redistricting is 60–700 times cheaper than existing models. We analyze how the **bisect** manifest format satisfies state and federal administrative procedure act (APA) record-keeping requirements, including the factual record standard of *Motor Vehicle Manufacturers Association v. State Farm*, 463 U.S. 29 (1983). We provide a model procurement specification that state redistricting offices can use to contract for algorithmic redistricting services or to evaluate vendor proposals.

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1 Introduction

The most common objection to algorithmic redistricting from state administrators is not technical—it is fiscal. “How much will this cost?” is the first question from legislative budget offices and redistricting staff. The answer is important: if algorithmic redistricting is expensive or administratively complex, its cost-benefit case is much weaker than if it is cheap and straightforward.

The answer is that algorithmic redistricting using the `bisect` platform is extraordinarily inexpensive by comparison to any alternative. The algorithm runs on public data that costs nothing to obtain, on commodity computing hardware, and produces a verifiable output in minutes. The dominant costs are human—staff time for quality assurance, public engagement, legal review, and documentation—and these costs are substantially lower than the human costs of any alternative redistricting process.

This paper is addressed to state administrators, legislative staff, budget officers, and procurement specialists who need to understand what it actually costs to implement algorithmic redistricting, what administrative law requires in terms of record-keeping and procedural compliance, and how to write a procurement specification that accurately describes the services they need.

Section 2 provides a full cost accounting for a `bisect` run, breaking costs into data acquisition, compute, staff time, audit, and legal review components. Section 3 compares these costs against the Iowa model (the cheapest existing alternative) and the California model (the most expensive). Section 4 analyzes APA record-keeping requirements and how the `bisect` manifest satisfies them. Section 5 provides a model procurement specification. Section 6 concludes.

One important framing note: the relevant comparison is not “algorithmic redistricting cost” versus “zero,” but “algorithmic redistricting cost” versus “redistricting litigation cost.” States that have litigated their redistricting maps have spent \$1M to \$40M in legal fees per cycle, with no guarantee of a neutral outcome. The question is not whether algorithmic redistricting costs money—it does—but whether it costs more or less than the alternatives.

2 Full Cost Accounting for a `bisect` Run

We provide a full cost accounting for a complete `bisect` redistricting run for a typical state (14 congressional districts; approximately 3,000 census tracts). All costs are in 2026 dollars.

2.1 Data Acquisition

Census redistricting data (PL 94-171 file): Free. The Census Bureau publishes PL 94-171 data in a standard format for all 50 states. The `bisect` platform fetches this data automatically:

```
bisect fetch --year 2020 --workers 8
```

Total download volume: approximately 40GB for all 50 states. For a single state: 0.5–2GB depending on state size. Download time: 5–20 minutes on a broadband connection. Cost: \$0 (data charges) + staff time to initiate and monitor the download.

TIGER census tract boundary files: Free. The Census Bureau publishes TIGER/Line shapefiles for all geographic units. These are downloaded automatically as part of the `bisect fetch` process. Total volume for all states: approximately 3GB.

Adjacency graph construction: Automatic. The `bisect` platform constructs the tract adjacency graph from the TIGER files as part of the build process. No additional data acquisition is required.

Total data acquisition cost: \$0 in licensing fees. Staff time: approximately 2 hours to initiate, monitor, and verify the data fetch.

2.2 Compute

Hardware: The `bisect` platform runs on any modern x86-64 Linux or Windows machine. A cloud instance with 8 cores and 32GB RAM (e.g., AWS `c5.2xlarge` at \$0.34/hour in 2026 pricing) is sufficient for all 50 states.

Wall-clock runtime: A full 50-state congressional redistricting run on an 8-core machine completes in approximately 18 minutes:

```
bisect build congressional_2030 --year 2030 --workers 8
```

A single state completes in seconds to minutes depending on state size.

Cloud compute cost: 18 minutes at \$0.34/hour = \$0.10. For a full analysis pipeline including all analysis types and report generation:

```
bisect label-analyze congressional_2030 --year 2030 --types all
bisect label-report congressional_2030 --year 2030 --format html json
```

Total compute time: approximately 2 hours. Cost: \$0.68.

Storage: Output files for a 50-state run (district assignments, manifests, analysis reports): approximately 5GB. Cloud storage at \$0.023/GB/month for 12 years: approximately \$16.

Total compute cost: Under \$20.

2.3 Staff Time

Staff time is the dominant cost. We estimate the following components for a state redistricting office implementing algorithmic redistricting for the first time:

Table 1: Staff time estimate for first-cycle algorithmic redistricting implementation

Activity	FTE Fraction	Duration	Cost
Software installation and training	1.0	2 weeks	\$3,800
Data fetch and quality verification	0.25	2 weeks	\$950
Baseline run and output review	0.5	1 week	\$950
Public hearing preparation and facilitation	1.0	4 weeks	\$7,600
Departure documentation (if any)	0.5	2 weeks	\$1,900
Legal review of baseline and departures	(outside counsel)	3 weeks	\$15,000
VRA demographic analysis	(outside demographer)	3 weeks	\$12,000
Independent technical audit	(university or contractor)	2 weeks	\$8,000
Public website and documentation	0.5	2 weeks	\$1,900
Total (first cycle)			\$52,100

Staff cost assumptions: in-house GIS analyst at \$95,000/year loaded cost (\$1,900/week); outside counsel at \$5,000/week; outside demographer at \$4,000/week; technical audit contractor at \$4,000/week.

Subsequent cycles: Staff familiarity with the process and pre-established legal and audit relationships reduce repeat-cycle costs substantially. We estimate subsequent cycles at 50% of first-cycle cost (\$26,000).

2.4 Summary

Table 2: Total cost estimate for algorithmic redistricting (bisect platform)

Component	Cost
Data acquisition	\$0
Compute (cloud)	\$20
Storage (12-year retention)	\$20
Staff time (first cycle)	\$52,100
Total (first cycle)	\$52,140
Total (subsequent cycles)	\$26,000

The cost estimate in Table 2 reflects a competent, properly resourced implementation. States that already have GIS staff familiar with census data (most do) and existing relationships with election law counsel will spend less. States starting from scratch will spend

somewhat more. The range is approximately \$35,000–\$75,000 for a first-cycle implementation.

3 Comparison with Existing Redistricting Models

We compare the `bisect` cost estimate against two benchmarks: the Iowa Legislative Services Agency model and the California Citizens Redistricting Commission.

3.1 The Iowa Model: \$3 Million Per Cycle

Iowa’s Legislative Services Agency is widely regarded as the most cost-effective non-partisan redistricting model in the United States. Its 2021 redistricting cycle cost approximately \$3 million, including:

- LSA staff time (three full-time staff over 8 months): approximately \$480,000
- Outside counsel (constitutional and VRA review): approximately \$200,000
- Public hearings (statewide; multiple rounds of maps): approximately \$150,000
- Legislative review process (committee hearings, floor debate): approximately \$400,000
- Technical systems (GIS software, mapping tools): approximately \$75,000
- Document publication and public engagement: approximately \$50,000
- Expert demographic analysis: approximately \$120,000
- Contingency and administrative overhead: approximately \$1,525,000

The \$3 million Iowa estimate includes the cost of Iowa’s three-map process (the LSA draws and redraws if rejected) and the legislative review process. The dominant cost is staff time for a process that takes 8 months, compared to the `bisect` implementation’s 12 weeks.

The algorithmic approach eliminates most of the map-drawing staff time (the algorithm draws the baseline in minutes, not months) and reduces the number of maps drawn (one baseline plus documented modifications, not multiple redrafts). The legislative review process cost is unaffected (it is a function of legislative procedures, not algorithm design), but is computable only if the state retains a legislative review stage.

Iowa-versus-bisect comparison: \$3,000,000 versus \$52,000. The algorithmic approach costs approximately 1.7% of the Iowa model.

3.2 The California Model: \$35 Million+ Per Cycle

California’s Citizens Redistricting Commission spent approximately \$35 million on the 2021 redistricting cycle, including:

- Commissioner compensation and benefits (14 commissioners, 18 months): approximately \$5.3 million
- CCRC staff (executive director, legal counsel, GIS staff, administrative): approximately \$8.5 million
- Public engagement (60+ hearings, translation services, website, outreach): approximately \$6.2 million
- Mapping contractors and technology: approximately \$3.1 million
- Legal costs (constitutional review, VRA analysis, litigation defense): approximately \$7.5 million
- Expert witnesses and demographic analysis: approximately \$2.8 million
- Translation and accessibility compliance: approximately \$1.6 million

California’s high cost reflects its comprehensive process—60+ public hearings, extensive translation services, full-time commissioners for 18 months—and its constitutional entrenchment requirements. The CCRC’s maps were challenged in litigation, adding further legal costs.

California-versus-bisect comparison: \$35,000,000 versus \$52,000. The algorithmic approach costs approximately 0.15% of the California model.

3.3 Redistricting Litigation: The Hidden Cost

The cost comparison above excludes redistricting litigation costs, which are the dominant expense in states where maps are challenged. Table 3 summarizes publicly reported litigation costs in major redistricting cases since 2010.

No state using algorithmic redistricting with a verifiable manifest has faced redistricting litigation over map authenticity. The **bisect** manifest’s byte-reproducibility standard eliminates the “is this map neutral?” disputes that generate most redistricting litigation costs.

Table 3: Redistricting litigation costs in selected states, 2010–2024

State	Cost	Notes
North Carolina	\$37M+	Multiple cycles; LWV, NAACP cases
Pennsylvania	\$18M+	<i>LWV v. PA</i> ; Senate and House maps
Alabama	\$22M+	Two cycles; <i>Milligan</i> and predecessors
Maryland	\$12M+	6th district litigation
New York	\$15M+	<i>Harkenrider</i> ; multiple maps
Ohio	\$25M+	Multiple rounds of maps; seven Supreme Court orders
Wisconsin	\$11M+	Multiple cycles; legislative and congressional
Texas	\$35M+	DOJ, private litigation; three census cycles
8-state total	\$175M+	

3.4 Cost-Benefit Summary

The cost-benefit case for algorithmic redistricting is compelling on pure financial grounds:

- Algorithmic redistricting (\$50K) versus Iowa model (\$3M): $60\times$ cost reduction
- Algorithmic redistricting (\$50K) versus California model (\$35M): $700\times$ cost reduction
- Algorithmic redistricting (\$50K) versus redistricting litigation (\$10M–\$40M): $200\text{--}800\times$ cost reduction

Even if implementation costs are three times higher than estimated (due to state-specific complications, higher legal review needs, or first-cycle inefficiencies), the cost comparison is overwhelming. States that have spent double-digit millions on redistricting litigation could have funded algorithmic redistricting for 100 cycles with the same budget.

4 Administrative Law and APA Record-Keeping

State redistricting processes are quasi-legislative in most states: they are conducted by the legislature or by commissions that exercise delegated legislative authority. Some states, however, have treated redistricting commissions as administrative agencies subject to state administrative procedure act (APA) requirements. Federal redistricting under the DIA (P.1) would involve administrative action by the Census Bureau subject to the federal APA. This section analyzes how the `bisect` manifest format satisfies APA record-keeping requirements.

4.1 The APA Factual Record Standard

The federal APA requires that agency action be based on a “factual record” that supports the agency’s decision, and that the agency explain its reasoning in a manner reviewable by courts. *Motor Vehicle Manufacturers Association v. State Farm Mutual Automobile Insurance Co.*, 463 U.S. 29 (1983), established that courts must invalidate agency action that is “arbitrary, capricious, an abuse of discretion, or otherwise not in accordance with law” based on the agency’s own record. The agency must have “examine[d] the relevant data and articulate[d] a satisfactory explanation for its action including a rational connection between the facts found and the choice made.” *Id.* at 43.

Redistricting decisions—whether by a commission, by a court-appointed special master, or by an agency under a federal statute—must satisfy an analogous standard. The decision-maker must have examined the relevant data (population, geography, VRA demographic information), must have applied the stated criteria (population equality, compactness, political subdivision integrity, VRA compliance), and must explain departures from neutrality.

4.2 How the bisect Manifest Satisfies the Standard

The `bisect` manifest satisfies the APA factual record standard in three ways.

(1) Complete data provenance. The manifest records the specific data sources used—the census release identifier, the TIGER adjacency file SHA-256 hash, and the population source (total, VAP, or CVAP). Any reviewing court can verify that the decision-maker used the correct data by checking the manifest’s recorded hashes against the Census Bureau’s published hashes for the same releases. This establishes the completeness and accuracy of the factual record with mathematical certainty.

(2) Criteria application. The manifest records the algorithmic parameters (balance tolerance, edge-weight schema, seed) that implement the stated redistricting criteria (population equality within tolerance, compactness optimization). The manifest’s parameters are the criteria in quantitative form: there is a direct, auditable connection between the stated criteria and the recorded parameters.

(3) Departure documentation. The Commission Adjustment Protocol (P.3, §4) requires written Departure Findings for each modification of the algorithmic baseline. These findings are the “satisfactory explanation” the APA requires: they identify the specific data (VRA demographic data, community of interest evidence) that motivated each departure and the specific statutory criterion (VRA §2, political subdivision integrity) that authorized it. The difference file comparing the baseline to the final map quantifies the extent of each departure.

4.3 State APA Requirements

Most states have APAs that closely follow the federal model. The **bisect** manifest satisfies these requirements for the same reasons it satisfies the federal standard: data provenance, criteria application, and departure documentation are all recorded.

Some state APAs require that agency rules be subject to notice-and-comment rulemaking. If the adoption of algorithmic parameters (balance tolerance, edge-weight schema) constitutes “rulemaking” under the applicable state APA, the state redistricting authority must publish proposed parameters in the state register, accept public comment, and publish a final parameters document with responses to significant comments before the redistricting run. The **bisect** platform supports this process: parameters can be specified in a YAML configuration file that is published for comment:

```
# configs/state_congressional_2030.yml
name: state_congressional_2030
algorithm:
  structure: prime-factor
  weights: county
  search: convergence
  convergence_threshold: 600
  balance_tolerance: 0.5
workers: 8
years: ["2030"]
```

The YAML configuration file is a complete and human-readable specification of the algorithmic parameters, suitable for inclusion in a notice-and-comment rulemaking record.

4.4 12-Year Data Retention

APA record-keeping requirements typically require agency records to be retained for the period during which decisions can be challenged. Redistricting decisions are challengeable until the next redistricting cycle (10 years) plus the time for litigation to conclude (2–4 years). We recommend 12-year retention as a conservative standard covering one full redistricting cycle plus a full litigation window.

The **bisect** platform’s output artifacts—district assignments (**algorithm_output.json**), manifests (**manifest.json**), analysis reports, and departure documentation—together constitute the complete administrative record. These should be archived in a publicly accessible

location (state government website or state archives) for the required retention period. Storage costs for 12 years of retention are under \$20 (Section 2 cost accounting).

The DIA model statute requires 12-year data preservation explicitly (§105(e)), and states adopting algorithmic redistricting through other pathways should adopt equivalent retention periods through commission operating rules or statutory mandate.

5 Model Procurement Specification

We provide a model procurement specification that state redistricting offices can use to contract for algorithmic redistricting services or to evaluate vendor proposals. The specification is designed for a state with 10–20 congressional districts; it should be adapted for larger or smaller states.

5.1 Model Procurement Specification: Algorithmic Congressional Redistricting Services

SOLICITATION TITLE: Algorithmic Congressional Redistricting Services, [Year] Decennial Census Cycle

ISSUING AUTHORITY: [State] Legislative Services Bureau / Redistricting Commission / [other appropriate body]

PROCUREMENT TYPE: Professional Services Contract

CONTRACT PERIOD: [Census data release date] through [18 months after data release]

ESTIMATED CONTRACT VALUE: \$35,000–\$75,000 (see Section 3 below)

Section 1: Background and Purpose

[State] [Commission/Legislature/Bureau] seeks to procure algorithmic redistricting services for the [Year] congressional redistricting cycle. The successful contractor will produce an algorithmic baseline map for [State]’s [N] congressional districts from the [Year] decennial census redistricting data, using a deterministic, partisan-blind algorithm whose output is publicly verifiable from recorded inputs.

The algorithmic baseline map will serve as the starting point for [State]’s redistricting process, subject to modifications required by the Voting Rights Act and

[State] constitutional or statutory requirements. The contractor’s deliverable is the baseline map and associated documentation, not the final redistricting plan.

Section 2: Required Deliverables

The contractor shall deliver, within [60] days of receiving the decennial census redistricting data release:

(a) Algorithmic Baseline Map. District assignments for each census tract in [State], assigning each tract to one of [N] congressional districts, with the following properties:

- (i) Population balance within 0.5% of ideal district population ($(\text{[State population]} / \text{[N]})$);
- (ii) Geographic contiguity of all districts;
- (iii) Compactness optimization using recursive bisection with METIS edge-cut minimization;
- (iv) No use of partisan registration data, election results, race, ethnicity, or incumbency information as algorithm inputs.

(b) Reproducibility Manifest. A machine-readable file (`manifest.json`) recording: contractor software version (identified by SHA-256 hash of the distributable binary); census data release identifier; TIGER adjacency file SHA-256 hash; population source; balance tolerance; seed; district count; and creation timestamp.

(c) Population Balance Certification. Output of the `bisect label-verify` command demonstrating that each district satisfies the population balance requirement.

(d) Analysis Report. A report including: population statistics by district; compactness scores (Polsby-Popper and Reock) by district; political subdivision split analysis (counties and municipalities split across district boundaries); and minority VAP percentages by district.

(e) Technical Documentation. Written documentation sufficient to allow [State] technical staff to independently reproduce the baseline map using the contractor’s software and the recorded manifest parameters.

(f) Training. Not fewer than 4 hours of training for [State] technical staff in operating the contractor’s software, interpreting the manifest, and running the verification command.

Section 3: Pricing Structure

Offerors shall provide firm-fixed prices for:

- (a) Baseline map production, manifest, and population certification: \$[bid amount]
 - (b) Analysis report: \$[bid amount]
 - (c) Technical documentation: \$[bid amount]
 - (d) Training (per hour, up to 8 hours): \$[bid amount per hour]
 - (e) Optional: Departure certification—re-running population balance after commission modifications (per modification): \$[bid amount per modification]
 - (f) Optional: VRA demographic analysis to inform modification decisions (per district): \$[bid amount per district]
-

Section 4: Minimum Technical Requirements

Offers will be evaluated only if the offeror’s proposed software satisfies all of the following requirements:

- (a) The software shall be open-source (source code publicly available) or shall include a provision allowing [State] to escrow the source code for independent audit.
- (b) The software shall produce a manifest satisfying the format requirements in Section 2(b).
- (c) The software shall include a verification command that any person can run to confirm that the recorded manifest parameters reproduce the published baseline map.
- (d) The software shall not require any input other than census redistricting data, TIGER boundary files, the target district count, the population balance tolerance, and a random seed. The software shall be incapable of accepting partisan registration data, election results, race, ethnicity, or incumbency data as inputs.

- (e) The software shall have been used to produce redistricting maps in at least one prior redistricting proceeding, or the offeror shall provide independent technical evaluation by a qualified academic institution confirming its suitability.

Section 5: Evaluation Criteria

Offers will be evaluated on: (1) Technical approach (40%): compliance with minimum requirements; clarity of verification process; quality of prior redistricting applications; (2) Price (40%): total price for deliverables (a)-(d) above; (3) Past performance (20%): references from prior redistricting applications; responsiveness to questions and modification requests.

Section 6: Data Rights

[State] shall retain ownership of all deliverables and shall have the right to publish all deliverables on its public website without restriction. The contractor shall retain no proprietary rights in the census data, TIGER data, or derived district assignments. The contractor's software and algorithms remain the contractor's property, subject to the escrow or open-source requirement in Section 4(a).

5.2 Notes for Procurement Officers

The minimum technical requirements in Section 4 are designed to ensure that the **bisect** platform (or any equivalent open-source platform) qualifies, while excluding closed-source tools whose internal operations cannot be independently verified.

The price structure in Section 3 allows states to scale the contract to their needs: small states with few congressional districts may need only deliverables (a)–(c); states with substantial minority populations may need the optional VRA demographic analysis.

The training requirement (4 hours minimum) is essential for ensuring that state technical staff can independently verify the baseline, operate the software for departure certification, and maintain the administrative record. Training costs should not be omitted from the budget.

6 Conclusion

The implementation costs of algorithmic redistricting using the **bisect** platform are a minor fraction of any alternative. At \$50,000–\$75,000 per cycle—versus \$3 million for the Iowa model and \$35 million for the California model—algorithmic redistricting is not merely cheaper but an order of magnitude cheaper. The comparison to redistricting litigation costs (\$10–\$40 million in contested states) makes the cost-benefit case overwhelming: the cost of implementing algorithmic redistricting is less than the legal fees for a single redistricting motion in major litigation.

The dominant cost driver is not the algorithm—compute costs are under \$20—but staff time for quality assurance, public engagement, and legal review. These costs are real and should not be minimized: algorithmic redistricting requires competent human oversight to verify outputs, document departures, and ensure VRA compliance. But these costs are substantially lower than in any alternative process because the algorithm eliminates the months-long map-drawing phase that generates the largest staff expenditures in traditional redistricting.

The **bisect** manifest format satisfies APA record-keeping requirements directly. The manifest provides complete data provenance (which census release and TIGER file were used), documents criteria application (which parameters implement which criteria), and enables departure documentation (the Commission Adjustment Protocol from P.3). The 12-year data retention requirement is achievable at negligible cost (\$20 in cloud storage) and satisfies the period during which redistricting decisions can be challenged.

The model procurement specification in Section 5 gives state redistricting offices a ready-to-adapt document for contracting algorithmic redistricting services. It establishes minimum technical requirements (open-source or escrowed code; reproducible manifest; partisan-blind algorithm) that protect states against vendors whose tools are technically unsound or manipulable. The evaluation criteria balance technical quality, price, and past performance in a way that favors well-documented, verifiable approaches.

For administrators and budget officers reading this paper: the question is not whether your state can afford algorithmic redistricting. The question is whether your state can afford the alternative. States that have spent double-digit millions on redistricting litigation could have funded algorithmic redistricting for 200 cycles. The budget case for algorithmic redistricting is as strong as the legal and political cases developed in P.0 through P.4. Together, the five case papers in Track P provide a complete implementation toolkit: the political economy analysis (P.0), the federal legislative template (P.1), the ballot initiative design (P.2), the commission integration protocol (P.3), the court order models (P.4), and the cost accounting

and procurement specification (P.5). The toolkit is ready. The political conditions for using it vary by state and by moment. But the tool itself is complete, operational, and affordable.

References